Laptop Price Predictor - Machine Learning Project

Project Overview

A machine learning web application that predicts laptop prices based on technical specifications. The model uses Random Forest Regression to estimate prices with **~79% accuracy**.

Key Features

- **Input Parameters**: RAM, Weight, Company, Type, OS, CPU, GPU, Touch Screen, IPS Display
- **Output**: Estimated price in LKR (Sri Lankan Rupees)
- **Interactive Web Interface**: User-friendly form with dropdowns and checkboxes
- **Real-time Predictions**: Instant price estimation

Technologies Used

- **Python** - Core programming language
- **Flask** - Web framework
- **Scikit-learn** - Machine learning (Random Forest Regressor)
- **Pandas/NumPy** - Data processing
- **Pickle** - Model serialization
- **HTML/CSS** - Frontend interface
- **Hugging Face Spaces** - Deployment platform

Data Processing Pipeline

1. **Data Cleaning**: Removed units (GB, kg) from RAM and Weight columns
2. **Feature Engineering**:
    - Created Touchscreen and IPS binary features from ScreenResolution
    - Categorized CPU brands (Intel i3/i5/i7, AMD, Other)
    - Categorized GPU brands (Intel, AMD, Nvidia)

    - Simplified OS categories (Windows, Mac, Linux, Other)
    - Grouped rare company brands into "Other"
3. **One-Hot Encoding**: Converted categorical variables to numerical format

Model Performance

- **Algorithm**: Random Forest Regressor
- **Accuracy**: ~79% (R^2 score)
- **Comparison with other models**:
    - Linear Regression: 73.4%
    - Lasso Regression: 73.1%
    - Decision Tree: 73.6%
    - Random Forest: **79.0%** (Best performer)

```
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
model_acc(lr)

from sklearn.linear_model import Lasso
lasso = Lasso()
model_acc(lasso)

from sklearn.tree import DecisionTreeRegressor
dt = DecisionTreeRegressor()
model_acc(dt)

from sklearn.ensemble import RandomForestRegressor
rf = RandomForestRegressor()
model_acc(rf)
```

```
LinearRegression() --> 0.7340672939512776
Lasso() --> 0.7306408763703351
DecisionTreeRegressor() --> 0.7362725809789635
RandomForestRegressor() --> 0.7904901641528917
```

```python
from sklearn.model_selection import  GridSearchCV

parameters = {'n_estimators':[10, 50, 100],
              'criterion':['squared_error', 'absolute_error', 'poisson']}

grid_obj = GridSearchCV(estimator =rf , param_grid= parameters)

grid_fit= grid_obj.fit(X_train, y_train)

best_model = grid_fit.best_estimator_
best_model
```

```
[170]:
RandomForestRegressor
▸ Parameters
```

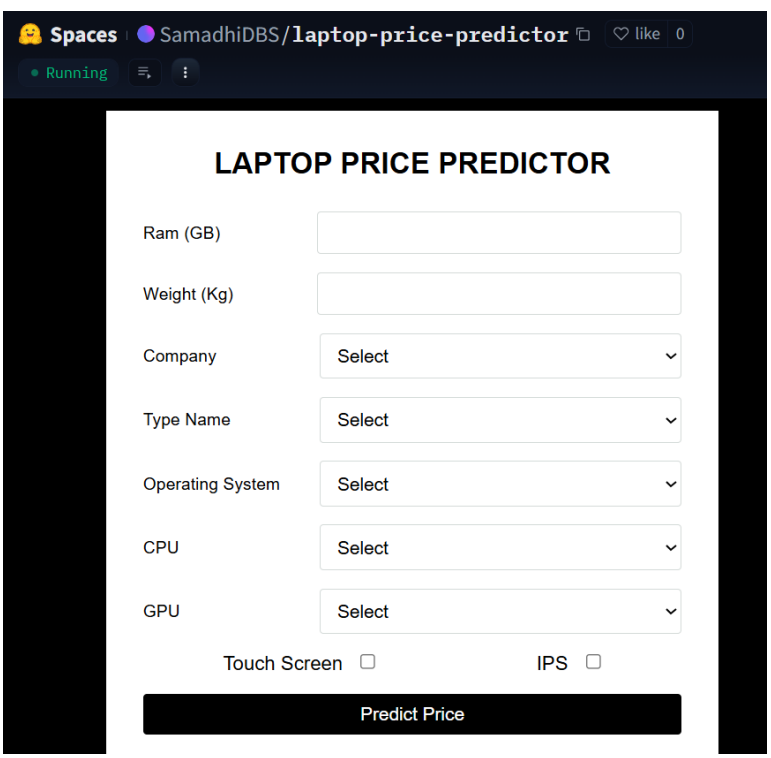
```
[171]:
best_model.score(X_test, y_test)
```

```
[171]:
0.768330276659335
```

Activate Window

Deployment

The application is deployed on Hugging Face Spaces and is publicly accessible:
https://huggingface.co/spaces/SamadhiDBS/laptop-price-predictor



Future Improvements

- Add more features (screen size, battery life, storage type)
- Improve UI/UX design
- Add data visualization
- Include confidence intervals for predictions
- Train on more recent laptop data

Author

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